

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 2$

(2) Find x-intercepts of $y = x^2 - 25$

(3) Find x-intercepts of $y = x^2 - 4$

(4) $y = -x^2$ opens:

(5) Find x-intercepts of $y = x^2 - 9$

(6) Find vertex of $y = x^2 + 14x + 1$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 8x + 2$

(2) Find vertex of $y = x^2 + 10x + 4$

(3) $y = -2x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 9$

(5) Find x-intercepts of $y = x^2 - 25$

(6) Find x-intercepts of $y = x^2 - 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 8$

(2) Find vertex of $y = x^2 + 14x + 8$

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 4x + 2$

(5) Find vertex of $y = x^2 + 16x + 8$

(6) Find vertex of $y = x^2 + 4x + 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 25$

(2) $y = -3x^2$ opens:

(3) $y = 3x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 9$

(5) Find x-intercepts of $y = x^2 - 4$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 2$

(2) Find x-intercepts of $y = x^2 - 9$

(3) Find x-intercepts of $y = x^2 - 16$

(4) Find x-intercepts of $y = x^2 - 25$

(5) Find vertex of $y = x^2 + 4x + 3$

(6) Find vertex of $y = x^2 + 16x + 5$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) $y = -3x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 16$

(4) Find vertex of $y = x^2 + 4x + 7$

(5) Find x-intercepts of $y = x^2 - 25$

(6) Find x-intercepts of $y = x^2 - 25$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 2x^2$ opens:

(2) Find x-intercepts of $y = x^2 - 9$

(3) Find vertex of $y = x^2 + 6x + 1$

(4) Find vertex of $y = x^2 + 4x + 7$

(5) $y = 2x^2$ opens:

(6) $y = 2x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) Find x-intercepts of $y = x^2 - 4$

(3) Find vertex of $y = x^2 + 16x + 1$

(4) Find vertex of $y = x^2 + 16x + 3$

(5) $y = -2x^2$ opens:

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 7$

(2) Find x-intercepts of $y = x^2 - 4$

(3) $y = -2x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 25$

(5) Find vertex of $y = x^2 + 10x + 8$

(6) Find vertex of $y = x^2 + 4x + 5$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) Find vertex of $y = x^2 + 10x + 4$

(3) Find x-intercepts of $y = x^2 - 16$

(4) $y = 2x^2$ opens:

(5) Find vertex of $y = x^2 + 16x + 9$

(6) $y = -3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 3$

(2) $y = 2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 16x + 8$

(5) Find x-intercepts of $y = x^2 - 9$

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) Find vertex of $y = x^2 + 14x + 9$

(3) $y = 3x^2$ opens:

(4) $y = 3x^2$ opens:

(5) Find vertex of $y = x^2 + 14x + 6$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 4$

(2) $y = 2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 10x + 8$

(5) $y = 2x^2$ opens:

(6) Find vertex of $y = x^2 + 16x + 8$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find vertex of $y = x^2 + 14x + 8$

(4) Find x-intercepts of $y = x^2 - 16$

(5) Find vertex of $y = x^2 + 16x + 4$

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) $y = 2x^2$ opens:

(3) $y = -x^2$ opens:

(4) $y = 2x^2$ opens:

(5) Find vertex of $y = x^2 + 4x + 4$

(6) Find vertex of $y = x^2 + 8x + 6$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) Find vertex of $y = x^2 + 10x + 1$

(3) Find vertex of $y = x^2 + 10x + 7$

(4) Find vertex of $y = x^2 + 12x + 9$

(5) Find vertex of $y = x^2 + 16x + 2$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 4$

(4) Find vertex of $y = x^2 + 6x + 8$

(5) Find vertex of $y = x^2 + 16x + 7$

(6) Find vertex of $y = x^2 + 8x + 4$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) $y = -2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 4$

(4) $y = -3x^2$ opens:

(5) $y = 2x^2$ opens:

(6) $y = 2x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -2x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find vertex of $y = x^2 + 14x + 3$

(4) Find vertex of $y = x^2 + 10x + 6$

(5) $y = 2x^2$ opens:

(6) Find x-intercepts of $y = x^2 - 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 4x + 7$

(2) Find vertex of $y = x^2 + 10x + 4$

(3) $y = -x^2$ opens:

(4) Find vertex of $y = x^2 + 16x + 7$

(5) Find vertex of $y = x^2 + 14x + 2$

(6) Find x-intercepts of $y = x^2 - 25$

Section 9 - Solutions

81a-1: $(-7, -47)$

81a-2: $x = \pm 5$

81a-3: $x = \pm 2$

81a-4: down

81a-5: $x = \pm 3$

81a-6: $(-7, -48)$

81b-1: $x = \pm 5$

81b-2: down

81b-3: up

81b-4: $x = \pm 3$

81b-5: $x = \pm 2$

81b-6: up

82a-1: $(-4, -14)$

82a-2: $(-5, -21)$

82a-3: down

Section 9 - Solutions (continued)

83b-1: $x = \pm 4$

83b-2: $x = \pm 2$

83b-3: $(-8, -63)$

83b-4: $(-8, -61)$

83b-5: down

83b-6: $x = \pm 4$

84a-1: $x = \pm 4$

84a-2: down

84a-3: $x = \pm 4$

84a-4: $(-2, 3)$

84a-5: $x = \pm 5$

84a-6: $x = \pm 5$

84b-1: up

84b-2: $x = \pm 3$

84b-3: $(-3, -8)$

Section 9 - Solutions (continued)

- 84b-4: $(-2, 3)$
- 84b-5: up
- 84b-6: up
- 85a-1: $(-6, -29)$
- 85a-2: $x = \pm 2$
- 85a-3: down
- 85a-4: $x = \pm 5$
- 85a-5: $(-5, -17)$
- 85a-6: $(-2, 1)$
- 85b-1: down
- 85b-2: $(-7, -40)$
- 85b-3: up
- 85b-4: up
- 85b-5: $(-7, -43)$
- 85b-6: up

Section 9 - Solutions (continued)

- 82a-4: $x = \pm 3$
- 82a-5: $x = \pm 5$
- 82a-6: $x = \pm 3$
- 82b-1: $(-6, -28)$
- 82b-2: $(-7, -41)$
- 82b-3: $x = \pm 3$
- 82b-4: $(-2, -2)$
- 82b-5: $(-8, -56)$
- 82b-6: $(-2, 5)$
- 83a-1: $(-7, -47)$
- 83a-2: $x = \pm 3$
- 83a-3: $x = \pm 4$
- 83a-4: $x = \pm 5$
- 83a-5: $(-2, -1)$
- 83a-6: $(-8, -59)$

Section 9 - Solutions (continued)

- 86a-1: up
- 86a-2: $(-5, -21)$
- 86a-3: $x = \pm 4$
- 86a-4: up
- 86a-5: $(-8, -55)$
- 86a-6: down
- 86b-1: $(-7, -46)$
- 86b-2: up
- 86b-3: $x = \pm 3$
- 86b-4: $(-8, -56)$
- 86b-5: $x = \pm 3$
- 86b-6: $x = \pm 4$
- 87a-1: $(-6, -32)$
- 87a-2: up
- 87a-3: $x = \pm 3$

Section 9 - Solutions (continued)

- 88b-1: up
- 88b-2: up
- 88b-3: down
- 88b-4: up
- 88b-5: $(-2, 0)$
- 88b-6: $(-4, -10)$
- 89a-1: down
- 89a-2: up
- 89a-3: $x = \pm 2$
- 89a-4: $(-3, -1)$
- 89a-5: $(-8, -57)$
- 89a-6: $(-4, -12)$
- 89b-1: $(-2, 3)$
- 89b-2: $(-5, -21)$
- 89b-3: down

Section 9 - Solutions (continued)

89b-4: $(-8, -57)$

89b-5: $(-7, -47)$

89b-6: $x = \pm 5$

90a-1: up

90a-2: down

90a-3: $x = \pm 2$

90a-4: down

90a-5: up

90a-6: up

90b-1: down

90b-2: up

90b-3: $(-7, -46)$

90b-4: $(-5, -19)$

90b-5: up

90b-6: $x = \pm 3$

Section 9 - Solutions (continued)

87a-4: $(-5, -17)$

87a-5: up

87a-6: $(-8, -56)$

87b-1: $x = \pm 4$

87b-2: $(-5, -24)$

87b-3: $(-5, -18)$

87b-4: $(-6, -27)$

87b-5: $(-8, -62)$

87b-6: up

88a-1: down

88a-2: up

88a-3: $(-7, -41)$

88a-4: $x = \pm 4$

88a-5: $(-8, -60)$

88a-6: $x = \pm 4$